

“I hope I never need one”: Unpacking Stigma in Aging in Place Technology

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ABSTRACT

Aging in place technologies are designed to extend independence and autonomy in older adults, promoting quality of life and peace of mind. Prior work has described how adoption and continued use of these technologies are low when they are perceived as reinforcing aging stigma or reminding older adults of negative aspects of aging. Building on past research, we investigate older adults' perceptions of four different kinds of aging in place technologies. Based on in-depth interviews with 18 older adults using a set of scenarios, we describe how different design characteristics contribute to perceived aging stigma. Additionally, we draw on Duner & Nordström's classification of coping strategies to discuss how specific kinds of technology can best support older adults during different stages of the aging process.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; • **Social and professional topics** → **Seniors**.

KEYWORDS

aging in place, stigma, older adults

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1 INTRODUCTION

The ongoing worldwide increase in the population of older adults [36] raises the challenge and opportunity regarding how we can use technology to support their well-being. Addressing existing challenges regarding how to support active aging and facilitate care is crucial for preparing for the future. The World Health Organization defines active aging as “the process of optimizing opportunities for health, participation and security in order to enhance quality of life as people age” [37]. Technology has a central part to play in this

process, as it can be designed to help older adults in their self-care efforts and to facilitate or mediate care from others.

Research on aging in place technologies typically focuses on promoting security by detecting or facilitating alerts in the case of an accident such as a fall (e.g., [33]). There are also systems available commercially with similar designs [32]. The rationale behind these systems is that they could allow older adults to live independently without compromising safety because a family member would be notified in case of an emergency. Therefore, moving in with family members just in case the older adult has a fall or other adverse health event (e.g., cardiac arrest) could be avoided or delayed.

However, the stigma associated with aging in place technology is a major sociotechnical issue that greatly limits adoption and effective use. Many older adults who could benefit from these systems reject them, often refusing to buy, wear, or use them when needed [15, 28, 35]. These issues are attributed to technology being seen or experienced as stigmatizing [25], dehumanizing [42], and a reminder of negative aspects of aging [40]. Although these issues have been consistently reported in prior work, there is little we know about how we can improve aging in place technology to reduce its association with aging stigma.

In this paper, we contribute to this effort through a qualitative study investigating older adults' perspectives towards four different kinds of technology designed to quickly respond to emergencies. We used scenarios illustrating active, passive, episodic, and continuous systems to understand how these different approaches compare in terms of perceived associated stigma. We find that check-in systems have, at the same time, the lowest stigma and lowest perceived utility. Home monitoring systems had the highest association with aging stigma. Lastly, while systems such as wearable alarm buttons were seen as most useful for providing security in case of an emergency, they were still substantially associated with aging stigma. Based on these results, we discuss how specific characteristics of aging in place technology design contribute to perceptions of stigma and what design approaches are most appropriate for older adults in different stages of aging.

2 RELATED WORK

Older adults predominantly prefer to remain in their homes [14]. Aging in place technology could allow them to do that safely for longer, extending their independence and quality of life.

2.1 Aging in place Technology

Aging in place technology broadly aims to support older adults as they age, delaying or removing the need to leave their homes to receive care from others [24]. There are different kinds of technologies that can be used for these purposes, such as Personal

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Emergency Response Systems, Monitoring, and Check in Systems. While the design of these technologies differ, they often include functionalities to detect or respond to emergencies.

Personal Emergency Response Systems (PERS) are currently the most widely known and adopted aging in place systems. A PERS typically consists of a wearable device with a button that can be used to call for help in case of an emergency, such as a fall. Getting help quickly is crucial because waiting for a long time to receive healthcare after a fall increases the risk of mortality and other adverse health outcomes [46]. Additionally, PERS can help address fear of falling, something that leads older adults to restrict their activities and may accelerate decline [25]. Although many PERS users self-rate as highly independent [32], typical users are most commonly described in the literature as frail women with physical impairments who live alone [50].

Despite their relative popularity, many issues that limit adoption, use, and effectiveness have been reported. For example, many who have acquired the device do not wear it or do not activate it when an emergency occurs [50]. PERS can lead to feelings of stigma, discrimination, dehumanization, lower self-efficacy, and increased work for older adults and caregivers [25, 42]. These devices also cause feelings of insecurity and reductions in activity instead of greater independence. For example, users may avoid leaving their homes due to fear of going outside the device radius [2].

Monitoring, such as using home sensors to detect changes in habits or well-being, is also a common approach for aging in place technology [8]. For instance, the Aware home system was used to monitor self-care among older adults with diabetes [21]. Keeping track of self-care tasks and activities of daily living such as cooking can help to increase awareness and serve as an incentive to act when there is room for improvement (e.g., remembering to take medications more consistently) [27]. Additionally, sharing monitoring data with family members can help to provide peace of mind [47]. However, this kind of system also can have a negative impact on users. Notably, even if it allows the older adult to remain home, a monitoring system changes the experience of being at home, since being monitored can represent giving up control [34].

Aging in place technology also includes check in systems. These systems are used daily, as people can press a button or send a message in the morning as a signal that they are doing well. Failing to send that signal would prompt a follow-up to see if the person needs help. For example, the Check in Tree system [1] was designed for a group of low socioeconomic status older adults to check on one another collaboratively. This approach can be used as alternatives to PERS, as they might be more accessible [25] and are less associated with aging stigma [5].

These technologies are particularly needed for older adults who have limited access to other assistance options, including those living in rural areas or with a low socioeconomic status [43, 54]. For example, older adults in rural areas might not have neighbors nearby whom they can ask for help in case of an emergency. They also face barriers related to lack of transportation and lack of local health services [19, 43]. Despite these challenges, rural older adults are underrepresented in prior research.

2.2 Coping with aging and care needs

Issues such as perceptions of aging in place systems and aging stigma are directly linked with the older adult's experience of aging and what kind of help they are willing to accept. Duner and Nordström [12] characterized three different strategies that older adults use to cope with gradual physical or cognitive decline: active, adaptive, and passive coping. While experiences vary, there is a tendency to move from active to passive over time.

Active coping involves great effort to maintain independence and compensate for limitations. There is a fear that slowing down too much could mean risking further decline. Eventually, this strategy becomes overwhelming and older adults have to surrender some responsibility. The adaptive coping strategy involves accepting help while having control over one's circumstances. In this case, older adults seek the help that they feel they need and advocate for themselves. They can push back against suggestions to relinquish control, such as moving to a nursing home. Lastly, passive coping is described as a complete surrender of control to others. At this point, older adults have passed a "turning point" where they give up their independence and need extensive care [12].

As they experience changes in their health and well-being, older adults may respond to specific issues in different ways. Problem-focused coping involves acting to address a specific issue, such as getting a walker due to walking difficulty. Emotion-focused coping is a psychological process of reinterpreting circumstances positively. Denial represents a barrier to addressing their difficulties [12], since a person might underestimate their need for help and exceed what is reasonable or acceptable.

These different aspects of the aging experience impact older adults' perceptions of aging in place technology. For example, they are more willing to use systems when they see themselves as more vulnerable [52]. At the same time, they fear that these tools would lead them to be perceived as frail [6]. These conflicting factors have been described as a "tug of war" between their previous self-identity and coping with risks of severe falls [25]. Those who opt not to use aging in place technology might instead compensate, for example, by reducing their life-space mobility to manage fall risks [39].

The negative associations of aging in place technology with stigmatized aging are one of the main issues facing researchers and designers in the field of Human-Computer Interaction (HCI) and aging [24]. In addition to resistance to adopt a system and having negative experiences, reinforcing negative perceptions of aging has measurable detrimental impacts on older adults. These systems can reduce self-efficacy, an important factor for successful aging [17], and increase work, both for older adults and for caregivers [42]. Negative reminders of health deterioration can increase feelings of anxiety about their future [6] and lead to worse outcomes, including increased risk of mortality [17].

Older adults can perceive different designs as more or less stigmatizing than others [5]. Negative experiences could be attributed to focusing on limitations [26]. As an alternative, researchers have suggested focusing on supporting older adults' agency to manage their aging process [31] and focusing on comparisons with peers that highlight their strengths [24, 54]. However, these recommendations are difficult to implement in systems designed to provide help when a fall or other emergency happens. Factors such as how

emergencies are detected or communicated to users and how often users interact with the system might influence stigma.

Therefore, there is a need for further research to investigate how we can design aging in place systems that provide benefits such as increased safety and peace of mind while minimizing aging stigma [6]. In this paper, we contribute to this aim by studying how older adults perceive different design approaches for aging in place technology. We conducted a qualitative study to evaluate perceptions among older adults who do not yet use any aging in place technologies since issues related to perceived need and stigma are present both pre- and post-adoption [38].

3 METHODS

We conducted in-depth semi-structured interviews with 18 older adults using four different scenarios to discuss multiple kinds of aging in place technology. This research project was reviewed and approved by the institution's IRB.

3.1 Technologies and Scenarios

We created scenarios to provide participants with different examples of technology, with the goal of prompting discussions during the interviews. These scenarios illustrated four kinds of aging in place technology with different characteristics: active or passive and episodic or continuous (Table 1). Active systems require some form of interaction from users, such as pushing a button to call for help, while passive systems use sensors to automatically detect an accident or emergency. Episodic systems are only activated when an accident or fall occurs, while continuous (or recurring) systems are active at least daily. We chose these characteristics to differentiate between common aging in place approaches to emergency response and user interaction. However, they are not meant to be comprehensive classifications of technologies for aging in place. To the best of our knowledge, there is no existing alternative that could have been applied in the study. This classification would require validation before it is used for other purposes in future research. However, such validation falls outside the scope of this paper.

There were four scenarios in total, each one corresponding to one of the four kinds of technology. The scenarios for each technology were purposefully very similar, illustrating the technology in the context of detecting or responding to an emergency. For example, in the Check in system scenario, the daughter character contacts the older adult after they have not "checked in" for the day (Figure 1). In the Home monitoring scenario, the daughter character receives a notification on her phone indicating unusual activity on that day. This kind of use case was chosen because it was applicable to the four different kinds of technology. We aimed to reduce as much as possible any differences in use cases among the different scenarios so that any differences in the results could be attributed to the technology design itself. Similar to prior work, we used the scenarios to depict the use of different technologies with similar purposes, aiming to understand the perspectives of older adults about each design [7]. The systems are:

PERS (active, episodic): a wrist-worn wearable with a button that can be pressed to call for help in an emergency (e.g., fall). In the scenario, the person who is notified is a friend who lives nearby.

Table 1: Aging in place technologies used in this study, by use frequency and level of interaction.

	Active		Passive
Episodic	Personal	Emergency	Fall Detection device
	Response System (PERS)		
Continuous	Check in system		Home monitoring system

Fall detection (passive, episodic): a wrist-worn wearable that automatically detects a fall and calls for help. Presented in the same scenario as the PERS.

Check in system (active, continuous): a device that can be used within a group so all can check-in on others (Figure 1). There is a binary interface that turns *on* when the person pushes the button each day, representing a signal that the person is doing well. In the scenario, the older adult uses the device with three family members.

Home monitoring (passive, continuous): a home-based system with motion sensors used to monitor a person's activity inside their home. It can detect changes in behavior in the short-term, in the case of an accident, and long-term, such as gradual reductions in movement. In the scenario, an adult child of the older adult receives a notification that the system detected unusually low activity.

3.2 Participants

Most participants (N=16) were recruited through research volunteer platforms.¹ These platforms are volunteer registries designed to facilitate recruitment for research studies. Researchers register the study on a website and can contact registered volunteers who match the inclusion criteria. Potential participants are also able to contact the researchers directly. Additionally, two people had participated in a past study about a different topic and had agreed to be re-contacted. The inclusion criteria were the following: being 65 or older, living in the community (i.e., not in a nursing home or assisted living), and living in a rural area or small town (i.e., less than 100,000 people). We chose these criteria because older adults living in rural areas are an underrepresented group in past work. Some research findings might not generalize to different populations. Additionally, all participants lived in the same state in the U.S. Midwest (Indiana). Participants' ages ranged from 66 to 84 (median = 69). Most of the participants were female (N=12) and the remaining were male (N=6). All but two participants were retired and most of them accessed the internet daily through a smartphone, tablet, or computer (Table 2).

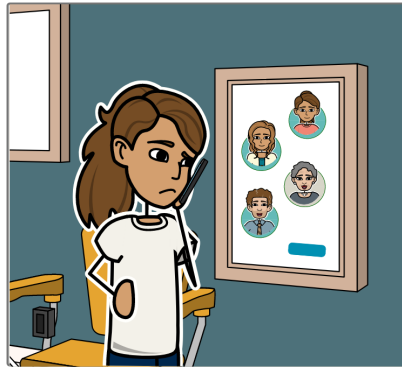
3.3 Data collection

Interviews were semi-structured and primarily focused on discussing aging in place technologies through scenarios. Participants were shown illustrations for each scenario (e.g., Figure 1). At the same time, the interviewer verbally explained the scenario by narrating it as a short story. Each kind of technology was discussed

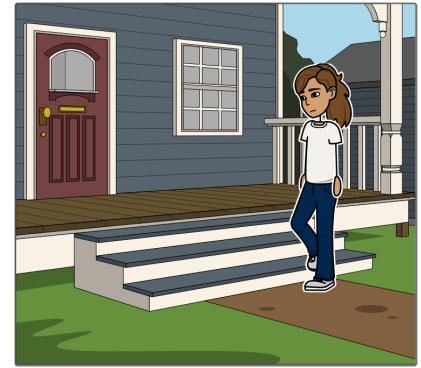
¹Indiana's All IN for Health (<https://allinforhealth.info>) and ResearchMatch (<https://www.researchmatch.org/>)



(a) Older adult checking in on a regular day by pressing a button.



(b) Daughter calling older adult who has not checked in.



(c) Daughter visiting the older adult in person after calling.

Figure 1: Scenario illustrating the Check In system in the study. The scenarios of other systems followed the same style and structure.

Table 2: Study participants

P	Age	Gender	Internet use	Race/ Ethnicity	Living
1	73	woman	daily	white	alone
2	75	woman	daily	white	with spouse
3	70	woman	weekly	white	alone
4	68	woman	daily	white	with spouse
5	66	man	weekly	hispanic/ latino	with daughter
6	69	woman	daily	white	with partner
7	68	woman	daily	white	alone
8	84	man	rarely	white	with spouse
9	67	woman	weekly	white	with spouse
10	72	woman	daily	white, hispanic/ latino	alone
11	66	man	daily	white	with spouse
12	66	woman	daily	white	alone
13	81	man	daily	white	with spouse
14	78	woman	weekly	white	alone
15	66	man	weekly	white	alone
16	66	woman	daily	white	with spouse, grandchild
17	72	man	weekly	white	with spouse
18	71	woman	daily	white	alone

separately, including questions around for whom it was suitable, any concerns around its use, and whether the participant would use it. Participants were asked to compare these devices at the end of the interview.

During the interview, participants were also asked about aging and adverse incidents (e.g., falls) among people they knew. Most shared stories about people they knew who experienced an accident

and could not call for help and other relevant situations. The interviews were conducted remotely using teleconferencing software and were audio-recorded.

3.4 Data analysis

We followed the thematic analysis method to analyze the interview data [4]. Interviews were transcribed and analyzed using an inductive approach. First, two researchers read the transcripts to familiarize themselves with the data. The researchers separately coded two interviews to generate initial codes. Then, the researchers discussed this initial step to examine what preliminary codes had emerged and combined them into a single codebook. These two researchers used the codebook to code the remaining transcripts, revising the codes iteratively. In this step, the first author coded most of the data and supervised the coding done by the second author. During the analysis process, the researchers searched for themes in the data by reviewing the codes and organizing them into higher-level codes. All authors discussed preliminary themes in several synchronous meetings. These themes were revised iteratively and organized into high-level themes. We present the results of this analysis in the next section, including interview quotes from different themes.

4 RESULTS

When discussing aging in place technologies, older adults had a binary perspective around who should or should not use them. Rather than seeing these tools as resources for the prevention of issues that could arise over a gradual aging process, they described how their purpose should be to address existing problems. Therefore, they described potential users narrowly, as very frail older adults with cognitive impairments who live alone. At the same time, they also discussed examples of older adults with these characteristics who were reluctant to use aging in place technology. For themselves, participants were more open to general purpose devices (e.g., a smartwatch) than specific aging in place technology, even when they could be used similarly.

4.1 Target users of aging in place technology: frail, living alone

Aging in place technologies were seen by study participants as geared towards people who already needed assistance due to frailty, having frequent falls, being confused, or having dementia. Additionally, they highlighted how all of the systems included in the study would only be needed if the person lived alone. For themselves, as older adults who did not have substantial cognitive or physical issues, they generally described themselves as not needing this kind of system yet.

Participants consistently saw aging in place technologies as tools for older adults who have substantial physical or cognitive impairments. For example, P10 described who should consider using such systems, due to increased risks:

"If you have any mobility issues, if you're using any assistive devices, if you're using a cane, something like that. I guess even if you have some mental confusion, and if you wander off, you're putting yourself more at risk of falling." (P10)

In addition to physical and cognitive impairments, participants also described how these devices would only be useful for people who live alone. Often, these two factors were mentioned together, as illustrated in the following quote:

"I think people, not only who live alone, but who have already demonstrated some degree of frail illness. If they need to have a walker, if they have a cane, if they have perhaps Parkinson's, which if they've had previous heart attacks or strokes. A pre-existing condition, I guess you'd call it." (P14)

Similarly, to P14, participants generally expressed how they did not believe that the aging in place technologies from this study would be useful for people who only fit one of these two criteria. For them, people who had others living in the same house would already be monitored and they should be able to call for help when needed without using these tools.

At the same time, a few participants showed uncertainty about whether similar systems would be useful for people with substantial impairments, given that they should need more monitoring and assistance than these designs would be able to provide. For example, P15 said that people who have cognitive impairments would not gain many benefits from a check-in system because they would not be living alone:

"Most of the time, people that have dementia or Alzheimer's are living with someone or in a facility. A lot of times, they're not by themselves anyway." (P15)

Similarly, P12 believed that the home monitoring system would be too intrusive for more independent people, but that it did not do enough for people who could benefit from being monitored continuously:

"If you get to the point where you need that much looking after, you might need more care. That's my opinion." (P12)

As they lived independently and mostly did not have substantial impairments, study participants expressed that they would not consider using one of the aging in place systems. Often, they explained

that they were in good health and were not concerned about having an accident such as a severe fall. For instance, P10 discussed how she had a low risk of having a broken bone due to a fall:

"I have my bone density tests, and my doctor has said my bones aren't going to break. They're strong. I don't feel like I'm at risk. I don't feel like I fit into that risk category." (P10)

Several people also talked about hoping to not get to a place of enough impairment to need this kind of assistance. P16 took this stance, emphasizing that she saw monitoring and emergency systems as something for other people:

"I hope I never need one. They're fine for people that feel they need one. It's a convenience and safety measure for them." (P16)

Like P16, most participants portrayed themselves as being separate and very much distinct from the target users of aging in place systems. However, this perception was not present when they discussed general purpose systems, such as smartphones and smartwatches.

4.2 General purpose systems: better accepted by active older adults

In comparison with the specific aging in place devices that were brought up during the interviews, several participants explained how they would prefer to use more general systems that could serve similar purposes. For example, instead of a specific wrist-worn PERS device for older adults, they mentioned how they would prefer a regular smartwatch that could also be used to contact others in case of an emergency. This kind of alternative was not mentioned to them by the researcher. Instead, participants spontaneously suggested it as an alternative.

When asked about their own potential use of aging in place systems, a few participants justified not needing them due to using alternative means of communication and services. Typically, these alternatives were general purpose systems or services. For example, P11 explained that he would not need an additional device to call for help because the phone would serve that purpose:

"I've got a phone with me all the time. I essentially have a de facto device." (P11)

We found a similar difference in their perspectives when they discussed these kinds of technologies with their younger close relatives. A few participants had received devices from relatives or had had conversations with them about such devices. Even when they were acquired for the same purpose, participants were more accepting of general devices, i.e., tools that are not specific for aging in place.

For example, P17 received a smartwatch as a gift from his son and daughter-in-law with the purpose to serve as a tool to call for help in an emergency. P17 explained:

"My daughter-in-law, they got me the Apple Watch. She said, if I'm out walking and running all by myself, I needed to have something to be able to make phone calls, so it works. I can get phone calls, or I can call somebody if I need to. I haven't really used that feature, but it's possible." (P17)

While P17 did not find that he needed a device to call for help, he used the smartwatch and recognized that it would work for that purpose. In contrast, participants who had been asked to use aging in place devices showed resistance to them. For example, P12 said that she would prefer to wear a smartwatch that could be used to call for help in the case of an emergency than a PERS that would be worn around her neck:

“My daughter has been wanting me to get one of those [PERS] and I don’t want to. [...] I am resistant to wearing a thing around my neck like that. [The smartwatch] is better, much better. [...] She knows that I get really dizzy because of my Meniere’s. I’ve been deaf in my left ear for 20 years or so. When I have a bad bout, the dizziness is overwhelming. She knows I have a bad back. [...] I just say, ‘No,’ and then she asks why, and I say, ‘No, I don’t want it, period.’ ” (P12)

These two examples also illustrate how the older adults’ adult children perceive a need for aging in place technologies such as PERS earlier than older adults themselves do. Similarly, participants consistently showed more interest in aging in place technologies when discussing them with a caregiver perspective, rather than the perspective of a care receiver.

When talking about the future, specifically when and if they would eventually need caregiving or assistance, a few participants also discussed general purpose communication devices rather than aging in place technologies. They envisioned services that would compensate for limitations such as difficulty cooking or driving, but would not be targeted *specifically* towards older adults. For instance, P10 gave examples of meal delivery services and autonomous cars as resources that could be useful for her in the future:

“I think by the time I can’t drive, maybe I’ll have drones delivering things or something. Or a car that drives itself. That’s what’s going to happen [...] I’m thinking technology will be able to help me out when I have to give things up. [...] I guess when the time comes, I don’t know if I would use them or if I would use Instacart or if anybody would even drive out here where I live, use Amazon to deliver whatever, to deliver my food.” (P10)

In summary, when discussing aging in place technologies, participants showed a preference for general purpose systems. Even though these devices could be used for similar purposes, such as getting help in case of emergency, they were more open to technology that is not specifically for aging in place.

4.3 Resistance from frail older adults: aging stigma as a barrier to technology use

There were various examples of resistance to aging in place technologies even from people who, according to the participants, did fit the criteria of possibly needing these systems. In this case, older adults talked about people they knew who were older than them, including relatives, friends, and acquaintances.

Several participants discussed PERS systems from the perspectives of caregivers since they had thought about their usefulness for

a parent or other close relative. For example, P7 described resistance from her mother in using tools such as a PERS:

“I tried to get my mother to use one [PERS], but she wasn’t having it. [...] She was stubborn. She didn’t like anything new or any changes. She lived with me. I was there a good portion of the time, but I worked. I was always worried something would happen when I wasn’t home. I know one night, she had gotten a new mattress and had trouble getting off of it, and subsequently knelt to the floor and couldn’t get up. [...] She had been calling for help and I didn’t know it. She couldn’t get herself up. She was crawling to another surface to find out something to hold on to. Had she had something like that, maybe we would’ve been notified sooner. It had a happy ending. We got her up. I said, ‘You need...’ [laughs] She said, ‘I don’t need anything like that.’ ” (P7)

Similarly, P16 recounted how their mother resisted wearing a PERS device. She explained that using this kind of device would mean admitting being frail and needing help:

“I remember when my sister bought one of those alert necklaces for my mother. It was extremely expensive. This was probably 20 years ago. My mother refused to wear it. [...] I think she felt that it was giving up her independence. [...] [using PERS] It’s saying, ‘Yeah, I am old. I may fall down. I may hurt myself. I don’t want to be there.’ ” (P16)

Such resistance is not restricted to digital technologies. Participants described how older adults often refuse to use other assistive tools, even when they could benefit from them. P1 explained how she observed this kind of resistance among some of her friends and acquaintances:

“They say, ‘I’m not ready for that yet.’ It’s the same way with canes, ‘I’m not ready to use a cane yet.’ That’s really frustrating. There’s a lot of reluctance to say, ‘I’m not capable of doing it myself.’ When I walk, I use a walking stick. I don’t use a cane, but those two walking sticks. Sometimes, I use one or two. People I know that are in worse shape than me will not use anything. It’s frustrating.” (P1)

A few participants had gone through experiences of trying to intervene in situations where older adults showed resistance, despite needing help. In these situations, participants discussed being concerned about their safety. At the same time, trying to persuade someone to seek or adopt some kind of assistance was a delicate process.

For example, P10 had worked as a volunteer for an agency that provided services to older adults in the local community. Her tasks included visiting their homes. She described what she could do when she noticed that the person needed more help than they were getting:

“If you didn’t feel that it was safe for them to be standing up in the kitchen, cooking, and doing that kind of stuff, try and persuade them to use different services. [...] You start out suggesting things. We would have

friendly visitors who would go over to check in. Some people liked that, and some people didn't want other people around. [...] Ultimately, if they needed a nursing home level of care, then help them realize with whatever family they would have, with their physician, etc., that it's time to get more help. It's a process. It's individual for everybody." (P10)

At the time of the interview, P4 was going through a similar process with her parent. She relied on others, such as a healthcare provider, as individuals that could help persuade the person who needs more care or assistance. P4 described her concerns around tending for her mother's safety without going against her wishes:

"I'm hoping the doctor will be [...] the bad guy and say that my mom can no longer drive. I can't be the bad guy. [...] She's always said, 'I don't wanna go live with you and your husband,' because she says it's not fair to our marriage. Then, this trip she said, 'If you make me move somewhere,' she says, 'I'll just give it up.' It's tough." (P4)

Although P16 described herself as a very active and healthy older adult, she described how she could empathize with more frail people who struggled with agreeing to receive help or supervision, such as her mother. She explained how needing to cope with health issues when it is not possible to treat and resolve them is frustrating and difficult to admit:

"At my age, where I do have some arthritis, and I do have a certain amount of pain, and when I go to get help for it, I'm told, 'Sorry, there's nothing we can do for you.' We're the young person inside, but we're the old person outside. We want to be that young person on the outside too, but we don't get there. It's gone. It's not fair. It's frustrating and depressing because you can't do what you want to do." (P16)

In this quote, P16 also explicitly addressed a theme of self-identity that was implicit in many of the findings. Overall, participants associated aging in place technologies as being for a specific kind of person, the frail older adult. To them, using these systems represents a shift in self-identity from active older adult to frail older adult.

4.4 Specific features for specific needs: limit assistance to what is necessary

While discussing each technology from the study, most participants framed them as addressing specific needs. Rather than seeing these different devices as complementary parts of a larger aging in place ecosystem, they assigned different features for people who might have particular kinds of impairments, who are at risk for falls or some other kind of accident. In other words, they held the perspective that an older adult would use different features or systems depending on their specific impairments, needs, and general preferences. These differences are shown in Table 3.

Most participants considered both episodic systems (PERS and fall detection) as the best options to get timely help in case of an emergency. They saw both systems as similar. Depending on what specific risks they face, they would find either an active (e.g., falls due to balance issues) or passive (e.g., risk of fainting) system more

appropriate for the person. For them, automatic fall detection would be particularly useful for incidents that left the person unconscious or impaired in some way that would make pressing the PERS button difficult (e.g., confused after a stroke or with a broken arm). P5 expressed this perspective, mentioning an example from a close relative:

"If someone falls, it could render them unconscious or unable to move their arm. I have an elderly parent who fell and hurt herself, and she hurt her arm, and it was immobilized. It wasn't broken, but that detection would be extraordinarily helpful and could prevent further harm." (P5)

Additionally, at the same time that the wrist wearable option was received well, participants had concerns about the effort required to use and maintain the device, such as regularly charging it, needing to learn how to use a new device, and its resistance to damage from everyday activities (e.g., water exposure) or accidents (e.g., breaking on impact). For these reasons, participants overall found that episodic systems would be ideal for older adults who do not have cognitive impairments. P7 discussed how this maintenance work would be difficult in that case:

"If you've got somebody with memory issues, they may not remember to put the watch on, put the device on, or activate something." (P7)

P4 mentioned how her own parent was not able to use PERS due to having dementia, even though she was independent enough to live alone:

"I got one [PERS] for my mom a couple of years ago. [...] She could not figure out how to make it work, though, so I sent it back. She has some mild dementia." (P4)

Regarding the check in system, participants overall found it to have some of the same issues as episodic systems, as older adults with cognitive impairments could have trouble remembering to use it daily, causing false negatives. P2 described this concern as a limitation for the system:

"If I had dementia, [the check-in system] might not work, because I'll forget to do it." (P2)

Additionally, given that the check in would only happen once a day, participants found that this system would be much less helpful to get timely help in case of an emergency. As they explained, if something happened shortly after they checked in, there would be no way to signal for help until 24 hours later. P6 showed concern about this scenario:

"Sam could have had an incident. She could have had a stroke. She could have fall and broken something at 10:05 AM. She's not expected to check in until 10:00 again, 24-hours later." (P6)

Participants also focused on the social aspects of the check-in system, both in terms of benefits and regarding potential issues. Given the perceived lower effectiveness of this system for emergency purposes, participants instead considered it to be an alternative way to socialize, particularly for people who were less familiar with other communication tools, such as texting. P11 said:

Table 3: Summary of how each technology design was associated with specific circumstances and user needs.

Technology	Best for	User
PERS	Falls, if able to call for help	Semi-independent
Fall detection	Emergencies, if confused or unconscious	Semi-independent
Check in system	Social interactions, low risk monitoring	Low technology literacy
Home monitoring	Homebound, with cognitive impairments	Very frail

“If somebody’s alone, it gives them more socialization. I think it’s very positive for them.” (P11)

In addition, a few participants also saw the check-in system as a tool that could facilitate socialization. For example, P17 explained that it could provide a signal indicating when would be a good time to call someone:

“It even helps if you want to socialize and stuff like that. You can at least know that they’re awake and you aren’t going to wake them up if they press the button. That would be good to have.” (P16)

At the same time, a few participants shared concerns about other users being too busy (e.g., with work) during the day to notice that someone has not checked in. If the older adult used the system with a group of younger close relatives, they might be busy and not check the system often enough. In that case, it would not work as a tool to get timely help, as P10 explained:

“If she lives in the same town and she’s free, she could drive over and see her mother. If she’s out of town or if she’s in the middle of whatever she does, surgery, [laughs] then it might be a little trickier to follow up. ‘Is this a medical emergency? Did mom forget to do this? Is mom off having fun somewhere and forgot?’ ” (P10)

Regarding the home monitoring system, participants found it to be one of the options most fitting for older adults with the most care needs, e.g., those who might have both cognitive impairments and physical impairments. They explained how its passive nature with a design that is not wearable makes it so there is no effort or learning required from the older adult, as shown in the following quote:

“You don’t have to do anything with [home monitoring], whereas the other ones, you actually have to do something. That might be helpful for people with dementia, early dementia. [...] If the watch had to be charged every night, I always charge my watch, I have a lot of friends, they don’t charge their cell phones.” (P1)

Participants saw both active systems as not appropriate for older adults with cognitive impairments. They also found that the episodic passive system would also require work, such as remembering to charge the device. Therefore, they explained that the home monitoring system would be the only feasible option.

At the same time, its continuous monitoring would limit privacy, making it an invasive system for more independent older adults. Most mentioned concerns about false positives in case the person is outside or is doing something that is out of the ordinary but benign,

such as sleeping in on a weekend. For example, P2 explained how it could make the older adult concerned about accidentally activating an alarm:

“I think it’s the invasion of privacy and the alarmist aspect, ‘Oh God, I sat down and I don’t usually sit down at 2 o’clock in the afternoon, so I’m going to alarm someone.’ I don’t know, don’t like it.” (P2)

In summary, participants found that each device had presented specific benefits and drawbacks, such that the best option would depend on the users’ needs and preferences. They discussed how certain systems would be more appropriate for people with high risk for falls, whereas others would be most useful for people with cognitive impairments.

These results illustrate older adults’ nuanced perspectives around aging in place technologies. Overall, they perceived these tools as being appropriate for others, and not them, even when their younger relatives disagreed. The association of aging in place with frail older adults with cognitive impairments highlights how the use of such systems is linked with their sense of self-identity.

5 DISCUSSION

This study revealed a binary perspective on aging in place technology, wherein older adults described potential users as frail, with cognitive impairments, and living alone. This perspective stands in contrast with designing for aging as a gradual process and with a preventative approach. Similarly, older adults have expressed in past research that they see aging in place technology as something for other people and not them, preferring non-medicalized technology for themselves (e.g., [10, 41]). According to a systematic review on barriers to adoption of aging in place technologies, stigma is discussed as a factor in 18% of articles, and a perceived lack of need is registered in 20% [55]. Other technologies intended for older adults, such as companion robots, also might be perceived as more detrimental to users’ dignity and sense of independence because of their design [7].

Our findings are consistent with prior research and provide a more detailed understanding of how specific design characteristics influence perceived stigma. By encouraging comparisons among aging in place technologies, we identify how their different characteristics impact their perceived usefulness, benefits, and drawbacks. Despite these differences, the very association of certain devices with frail older adults acts as a barrier for adoption. In this section, we discuss how different kinds of technology for aging in place align with different coping strategies, how these findings compare with related literature, and describe how these novel insights can contribute to future research and design in this space.

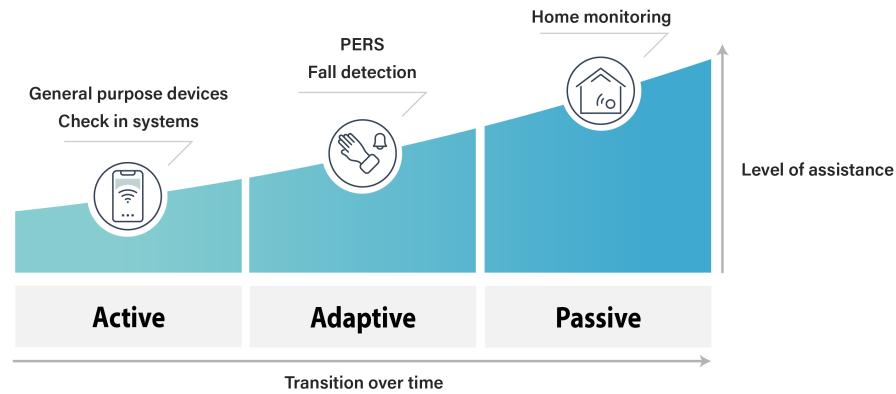


Figure 2: How specific aging in place technologies are aligned with different stages of aging.

5.1 Technology for different stages of aging

The systems included in the study were not deemed superior or inferior to others. Instead, participants discussed how they might serve different purposes. Based on these results, we can associate different kinds of aging in place technology with different coping strategies used by older adults: active, adaptive, and passive [12], as explained in section 2. These strategies are described as three parts of a continuum from independence to complete loss of autonomy that older adults potentially go through over time (Figure 2). Therefore, we can also conceive of them as different stages of the aging experience. We reflect on how different aging in place technology designs align with the three coping strategies below.

Older adults who are in the *active coping* stage push against the notion of facing frailty, as shown in section 4.3. This issue explains the reluctance in adopting technology for aging in place. For them, it would be most appropriate to use general purpose devices, such as smartwatches, for safety purposes (section 4.2). This approach does not consist of designing a smartwatch or app specifically for older adults that functions as a PERS. Rather, it involves providing features for a similar purpose in a system that is not geared towards any specific population. After all, people of any age may suffer severe falls and other accidents. While the suggestion of using general purpose technology to overcome stigma is not new (e.g., [16]), it seems more feasible for older adults now than it did in the past. This study illustrates how older adults have become more open to smart devices, making tools such as smartwatches more suitable as alternatives to PERS. Participants themselves mentioned these alternatives without prompting from researchers, a strong indicator that this option should be pursued. In addition to monitoring and emergency response, general technology such as widely adopted messaging apps can be used for intergenerational care [22, 30]. Check in systems would also fit best with people using active coping, perhaps close to the transition to adaptive coping. Still, the description of check in systems as providing social interaction as an alternative to texting or phone calls for older adults who do not text illustrates a preference for general purpose systems. Therefore, check in systems and other aging in place technologies might represent alternatives for older adults with lower technology literacy.

Older adults using *adaptive coping* are very deliberate about what they use. They need tools designed for specific impairments and concerns (section 4.4). For example, a person might be ready to use something for a specific need such as fear of falling, but not something more comprehensive that addresses several possible needs of a frail person. Consistent with this description, we find that episodic systems such as PERS and fall detection are seen as most useful for people who are semi-independent. Still, participants classified these two systems as fitting for people with different needs. For example, automatic fall detection was seen as most fitting for people who were at risk of being unconscious or otherwise incapacitated after a fall, including people at high risk for a stroke or fainting. Therefore, we argue that systems designed to address one specific need that older adults might face should be preferable to more comprehensive options. In addition to usefulness, this approach could be beneficial for addressing issues related to stigma and self-identity. A system designed for a very specific need would likely have less association with an overall idea of a frail and dependent older adult. For this reason, a person with such a need likely would be more open to using the system. Other researchers have argued that small specific solutions for autonomy are preferred in comparison with one-size-fits-all approaches [23]. This study provides additional evidence that supports this recommendation. Future research can support this design approach by studying older adults with specific impairments or needs, such as dementia [3, 11] or arthritis [20].

Lastly, older adults who use a *passive coping* strategy typically have given up their autonomy and need extensive care. In this study, the home monitoring system was perceived as closely aligned with the passive coping strategy. This kind of technology design is continuous and passive. Its passive nature is part of its classification as fitting for the frailest older adults, as participants explained that people with substantial physical or cognitive impairments would have difficulty properly using active systems. Similarly, the passive but episodic system was deemed as something that requires more effort because its wearable form factor would require work such as charging it and remembering to put it on. Additionally, the continuous nature of the home monitoring system was seen as addressing more than just emergencies, as something that monitors for more

behaviors and would detect other changes, such as spending more time resting.

Overall, extending independence is not aligned with the passive coping strategy. Therefore, designing technology for aging in place or active aging [37] can also be pursued by aiming to maximize periods of active and adaptive coping among older adults, delaying the transition into the period of passive coping. As argued in this section, using *specific technology that is aligned with the older adult's current coping strategy* is the best way to pursue this goal. These insights complement recommendations from prior work on more detailed aspects of technology design that contribute to stigma [29].

5.2 Aging in place and self-identity

The association of aging in place technologies with frailty and loss of autonomy illustrates how the intended benefit of these tools (i.e., extending independence) is undermined by how they are perceived. Similar to other care resources, they represent a decline, a shift in self-identity from independent to frail that older adults resist (sections 4.1 and 4.3). This binary perception reflects a bigger societal concept of aging as either good or bad, either normal or pathological [40]. Aging in place technology designed to promote the former is perceived by older adults as the latter. This association between aging in place technology and self-identity is consistent with prior work, where older adults described users of aging in place systems as frail and living alone [50], meaning that adopting them would also mean assuming these characteristics for themselves. Older adults who are perceived to need monitoring are no longer seen as equals by their peers, who might assess them as lacking mental acuity and social adequacy [9]. Stigma is also a barrier to the adoption of accessibility features because they are associated with disability. Even older adults who have accessibility needs (e.g., hearing loss) might not see themselves as disabled [18, 49], an issue also observed among people with mental health conditions [44]. Aging in place systems aim to improve quality of life by replacing other forms of care, but that is not enough to support older adults' sense of dignity and independence. In contrast, mainstream devices such as iPads can be perceived as status symbols, as using them demonstrates an ability to learn new things, something that is valued as a sign of cognitive competence in old age [53]. These self-identity factors can be more meaningful for older adults than specific details about how they receive care [31].

In this paper, we discuss different kinds of aging in place technology to surface how these differences impact how they are perceived and what features are most aligned with older adults' preferences. Although all systems were associated with aging stigma, we find that older adults perceived the check in system as the least effective system for addressing emergencies and the least appropriate option for people with cognitive impairments. Interestingly, prior work had found that check in systems have less stigma associated with them in comparison with PERS devices [5]. Perhaps the lower perceived usefulness of the check in system contributes to the lower association with stigma, due to a weaker attribution of its functionalities with frail individuals. Researchers have also reflected on the paradox regarding how technology designed to promote independence can instead create an experience of dependence on others [52]. Based on this example of the check in system, we can

surmise that there is a relationship between the degree of care or assistance provided by an aging in place system and how much it is perceived to be stigmatizing. This relationship makes sense intuitively because using a tool that provides a higher degree of assistance could imply that the person needs that much assistance. Hence the importance of avoiding implementing features with more assistance that the older adult needs or has opted for. Instead, using systems aligned with the coping strategy of the older adult, as discussed in section 5.1, should lead to fewer issues related to aging stigma.

Additionally, future research on aging in place technology should focus on the specific stage of aging corresponding to the technology in question. Researchers have raised issues related to considering older adults a homogeneous group, making their diversity invisible in HCI research [13, 45]. We contribute to research in this area by describing how different kinds of designs are most suited for different stages of the aging process. The insights provided in this paper can guide future research to create specific conceptualizations of who is the target population for a particular system and who should be recruited to participate in user studies and design workshops. Future work can build on this study by investigating how other kinds of aging in place designs fit into different stages of aging. Therefore, we recommend that future studies focus on specific coping stages and consider the needs and constraints associated with those stages.

5.3 Preventative technology for aging in place

Preventative systems would be most beneficial for older adults using active and adaptive coping strategies. Older adults in these contexts likely would refrain from using tools beyond what they already need (section 4.4). Ideally, older adults using each coping strategy should be able to use systems aligned with their needs at that stage. However, technology whose purpose is to detect or anticipate needs as they arise in the aging process cannot be limited to addressing older adults' existing needs. The perception that aging in place technologies are only for others, or for older adults who are frail and need a lot of care is a substantial barrier to the use of more preventative systems.

While the designs explored in this study were not flexible enough to adapt to different stages, proactive aging in place technology could become more flexible by increasing its acceptability. Because acceptability depends on how the system is perceived, we do not provide specific design implications regarding appropriate features for each stage. Devices with similar features, such as a wrist-worn PERS and a smartwatch with an emergency button, are perceived differently enough to belong in different stages. Nevertheless, based on the results, we can reflect on how to make progress towards more flexible aging in place technology. Acceptability due to perceived stigma is one-sided, as older adults reject technologies associated with subsequent coping stages. Concerns about suitability for the adaptive and passive stages depend, additionally, on whether the system provides enough assistance for the user without requiring unreasonable effort. To maintain acceptability in the active stage, mainstream devices such as smartphones and smartwatches could be used and adapted over time with the addition of software-based aging in place features. The required effort from older adult users

could be reduced over time without change in form factor, for example by installing long-lasting batteries or partially disabling resource-intensive software. Home monitoring systems for older adults could also start as mainstream technology and adapt over time. For example, sensors could be used primarily for home security for users in the active coping stage. Designing systems with enough flexibility to work in different coping stages could also reduce costs and user burden required for transitioning between very different systems over time.

We can also approach preventative technology for aging by incorporating some degree of monitoring into designs that are more aligned with older adults using active or adaptive coping. Secondary use of data from these systems is a possible direction towards this goal. For example, forgetting to use a check in systems or forgetting to take medications could serve as an early sign of cognitive decline. At the same time, it is necessary to inquire about to what extent older adults themselves want tools to detect early signs of physical or cognitive impairments. Given the challenging nature of coping with and responding to issues they already face, it is possible that detecting and noting early signs of aging-related decline would be more detrimental than not being aware of them. This might be a matter of framing or of focusing on early signs that are actionable, i.e., factors that they can respond to in order to prevent or delay a decline in health or loss of autonomy, as opposed to providing information that is interpreted as indicating a dire future.

5.4 Life transitions in aging

Many of the results from this study can be attributed to a mismatch between the self-identity of an older adult as independent and the perception of what kind of person should use aging in place technology. While participants saw themselves as self-sufficient and not needing external help, several discussed how they had family members who disagreed. Additionally, many participants also described knowing other older adults who resisted help despite needing it (section 4.3). Therefore, this mismatch is not only present between a potential older adult user and technology design, it also is part of their overall experience with the aging process, including their relationship with close family members.

As discussed in section 5.1, we can interpret these perceptions using the three different strategies used to cope with aging and receiving care. For example, older adults who are independent and use active coping likely would only use safety technology for general purposes, rather than specific aging in place technology. However, as they transition to adaptive coping, they might be willing to accept more specific systems for assistance. The process of transition between different strategies, including negotiations between different stakeholders, warrants further research. Such turning points [48], caused by major events such as diagnosis with a chronic illness or a stroke, impact family care dynamics.

Currently, aging in place systems are part of this transition process, but as a topic of disagreement between older adults and their family members. There is an opportunity to explore what it means to support people going through this delicate process. For instance, a caregiver sees aging in place tools as needed, but the older adult does not believe they have reached that point. Exploring how to

design to support both parties and to reconcile these two different perspectives could provide substantial benefit to people going through this stage of life. This dynamic is similar to the transition in health management responsibilities between parents and their children from childhood to young adulthood [51]. A tool designed for this kind of situation should facilitate negotiations and, at the same time, reduce frustrations and safety risks for older adults. It should focus on the psychological aspects of the transition, including reinterpreting their social roles and self-identities to come to terms with changes as they occur [12, 31]. This process might look very different for different transitions, such as moving from active to adaptive coping, in comparison with moving from adaptive to passive coping.

6 CONCLUSION

Stigma is a major barrier to the adoption and use of aging in place technology. In this paper, we present the results of a qualitative study investigating older adults' perceptions of different kinds of aging in place technology. We find that levels of perceived stigma vary, as certain designs are associated with a greater need for care and monitoring. Based on these findings, we describe what specific kinds of technology are most appropriate for each of the three stages of aging and discuss takeaways for future research and design in this area.

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